Exp No: 27

BINARY SEMAPHORE IN SYSTEMC

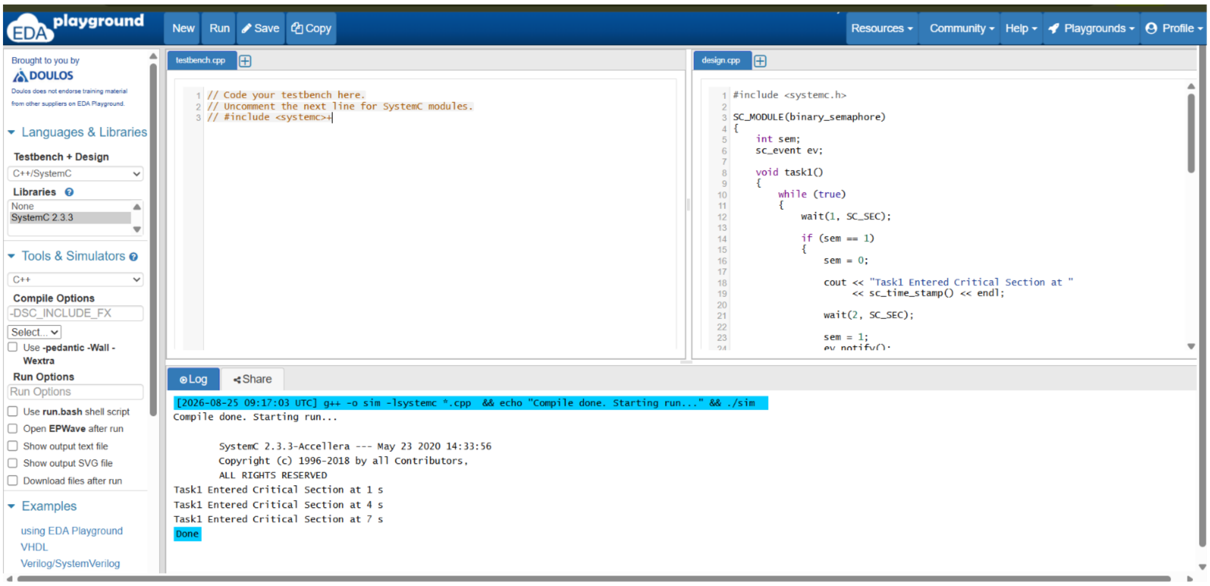
AIM:
To implement and simulate a Binary Semaphore using SystemC for task synchronization.

Program:

```
#include <systemc>
#include <tlm>
#include <tlm_utils/simple_initiator_socket.h>
#include <tlm_utils/simple_target_socket.h>

using namespace sc_core;
using namespace tlm;
using namespace std;

// Target
struct Target : sc_module
{
   tlm_utils::simple_target_socket<Target> socket;

   SC_CTOR(Target) : socket("socket")
   {
      socket.register_b_transport(this, &Target::b_transport);
   }

   void b_transport(tlm_generic_payload& trans, sc_time& delay)
   {
      int* data =
         reinterpret_cast<int*>(trans.get_data_ptr());

      cout << "Target received: " << *data << endl;

      delay += sc_time(10, SC_NS);
   }
};

// Initiator
struct Initiator : sc_module
{
   tlm_utils::simple_initiator_socket<Initiator> socket;

   SC_CTOR(Initiator) : socket("socket")
```

```
    {
        SC_THREAD(thread_process);
    }

    void thread_process()
    {
        tlm_generic_payload trans;

        sc_time delay = SC_ZERO_TIME;

        int data = 555;

        trans.set_data_ptr(
            reinterpret_cast<unsigned char*>(&data)
        );

        trans.set_data_length(sizeof(data));

        cout << "Initiator sending: " << data << endl;

        socket->b_transport(trans, delay);

        wait(delay);
    }
};

int sc_main(int argc, char* argv[])
{
    Initiator initiator("initiator");
    Target target("target");

    initiator.socket.bind(target.socket);

    sc_start();

    return 0;
```

}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Examples
using EDA Playground
VHDL
Verilog/SystemVerilog
testbench.cpp
// Code your testbench here.
// Uncomment the next line for SystemC modules.
// #include <systemc>
design.cpp
#include <systemc.h>
SC_MODULE(binary_semaphore)
{
int sem;
sc_event ev;
void task1()
{
while (true)
{
wait(1, SC_SEC);
if (sem == 1)
{
sem = 0;
cout << "Task1 Entered Critical Section at "
<< sc_time_stamp() << endl;
wait(2, SC_SEC);
sem = 1;
Log
Share
[2026-08-25 09:17:03 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...
SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
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Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s
Done